

FORMULAS AND IDENTITIES

Algebraic Square Formulae

1. $a^2 - b^2 = (a + b)(a - b)$

2. $(a + b)^2 = a^2 + b^2 + 2ab$

3. $(a - b)^2 = a^2 + b^2 - 2ab$

4. $ab = \frac{(a+b)^2 - (a^2 + b^2)}{2}$

5. $2(a^2 + b^2) = (a + b)^2 + (a - b)^2$

6. $4ab = (a + b)^2 - (a - b)^2$

7. $(a + b)^2 = (a - b)^2 + 4ab$

8. $(a - b)^2 = (a + b)^2 - 4ab$

9. $a^2 + b^2 = (a + b)^2 - 2ab = (a - b)^2 + 2ab$

Algebraic Cube Formulae

1. $(a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$

2. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

3. $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$

4. $a^3 + b^3 = (a + b)[(a + b)^2 - 3ab]$

5. $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$

8. $(a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$

9. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

10. $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$

11. $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$

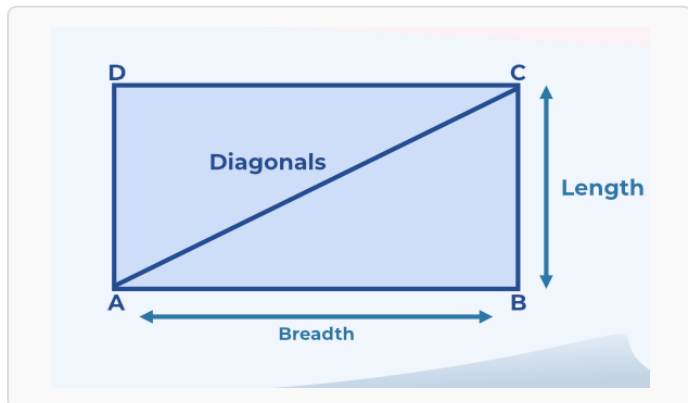
12. $a^3 - b^3 = (a - b)[(a - b)^2 + 3ab]$

Rectangle Formulae

1. Area: $A = l \times w$

2. Perimeter: $P = 2(l + w)$

3. Diagonal: $d = \sqrt{l^2 + w^2}$

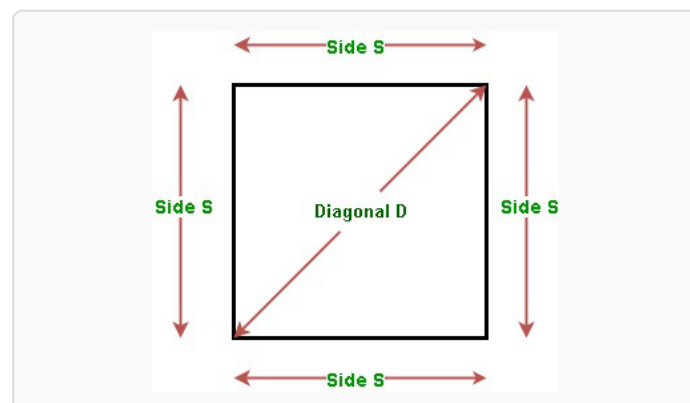


Square Formulae

1. Area: $A = a^2$

2. Perimeter: $P = 4a$

3. Diagonal: $d = a\sqrt{2}$

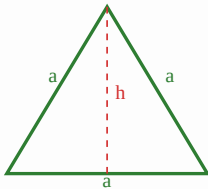


Equilateral Triangle

1. Height: $h = \frac{\sqrt{3}}{2}a$

2. Area: $A = \frac{\sqrt{3}}{4}a^2$

3. Perimeter: $P = 3a$

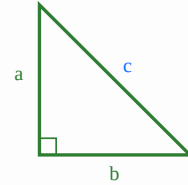


Right Angle Triangle

1. Area: $A = \frac{1}{2} \times b \times h$

2. Perimeter: $P = a + b + c$

3. Hypotenuse: $c = \sqrt{a^2 + b^2}$

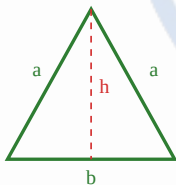


Isosceles Triangle

1. Height: $h = \sqrt{a^2 - \frac{b^2}{4}}$

2. Area: $A = \frac{b}{4} \sqrt{4a^2 - b^2}$

3. Perimeter: $P = 2a + b$

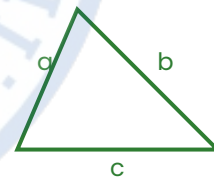


General Triangle (Heron's Formula)

Used when all three sides (a, b, c) are known.

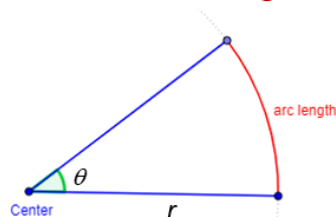
1. Semi-perimeter: $s = \frac{a+b+c}{2}$

2. Area: $A = \sqrt{s(s-a)(s-b)(s-c)}$



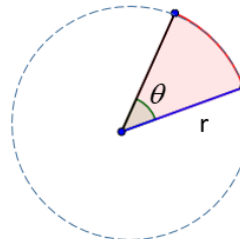
Circle Formulas

Arc Length and Area of Sector



If θ is measured in degrees then

$$\text{arc length} = \frac{\theta}{360^\circ} \times 2\pi r$$



If θ is measured in degrees then

$$\text{area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$